

BIOSTAT 724 - Homework 1

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Problem 1

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and $1 - p$ the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question, given that he or she answered it correctly? What happens when m is very large? What happens when p is close to 0?

Let's solve this using Baye's Theorem. First, it'll be helpful to define some terms:

- $P(\text{Knows}) = p$
- $P(\text{Guess}) = 1 - p$
- $P(\text{Correct} | \text{Guess}) = \frac{1}{m}$
- We want to know $P(\text{Knows} | \text{Correct})$

$$\begin{aligned} P(\text{Knows} | \text{Correct}) &= \frac{P(\text{Correct} | \text{Knows})P(\text{Knows})}{P(\text{Correct} | \text{Knows})P(\text{Knows}) + P(\text{Correct} | \text{Guess})P(\text{Guess})} \\ &= \frac{(1)(p)}{(1)(p) + (\frac{1}{m})(1 - p)} \\ &= \frac{p}{p + \frac{1-p}{m}} \end{aligned}$$

We can see from this result that as m gets arbitrarily large, the denominator approaches p and the probability goes to 1. This aligns with our intuition because a correct guess among a larger number of choices would be less likely. Likewise, if p is close to 0, the probability is also close to zero.

Problem 2

Suppose urn H is filled with 40% green balls and 60% red balls, and urn T is filled with 60% green balls and 40% red balls. Someone will flip a coin and then select a ball from urn H or urn T depending on whether the coin lands heads or tails, respectively. Let X be 1 or 0 if the coin lands heads or tails, and let Y be 1 or 0 if the ball is green or red.

Part a

Write out the joint distribution of X and Y in a table.

	X0	X1
Y0	0.2	0.3
Y1	0.3	0.2

Part b

Find $E(Y)$. What is the probability that the ball is green?

$E(Y) = \sum y \cdot f_Y(y)$. In our table, we see that Y can take on two values, 0 and 1. We can use the marginal probability to find that $\sum y \cdot f_Y(y) = (0)(0.5) + (1)(0.5) = 0.5$. Thus, $P(\text{Ball is green}) = 0.5$.

Part c

Find $\text{Var}(Y|X=0)$, $\text{Var}(Y|X=1)$ and $\text{Var}(Y)$. Thinking of variance as measuring uncertainty, explain intuitively why one of these variances is larger than the others.

Intuitively, I would expect the variance of Y without any additional information regarding X to be greater than either variances of Y given $X = 0$ and $X = 1$. That is, I expect $\max(\text{Var}(Y), \text{Var}(Y|X = 0), \text{Var}(Y|X = 1)) = \text{Var}(Y)$. My reasoning is that knowing something about X reduces our uncertainty of Y . Let's see if I'm correct.

$$\begin{aligned}\text{Var}(Y) &= \sum_{i=1}^n (Y_i - E[Y])^2 \cdot f(Y) \\ &= (0 - 0.5)^2(0.5) + (1 - 0.5)^2(0.5) \\ &= 0.25 \\ \text{Var}(Y|X = x) &= \sum_{i=1}^n (Y_i - E[Y|x])^2 \cdot f(y|x) \\ E(Y|X = x) &= \sum_{i=1}^n Y_i \cdot f(y|x) \\ E(Y|X = 0) &= (0)(0.4) + (1)(0.6) \\ &= 0.6 \\ E(Y|X = 1) &= (0)(0.6) + (1)(0.4) \\ &= 0.4 \\ \text{Var}(Y|X = 0) &= (0 - 0.6)^2(0.4) + (1 - 0.6)^2(0.6) \\ &= 0.24 \\ \text{Var}(Y|X = 1) &= (0 - 0.4)^2(0.6) + (1 - 0.4)^2(0.4) \\ &= 0.24\end{aligned}$$

As I guessed, knowing something about X does reduce the uncertainty around Y . We see that the conditional variances are both less than the variance of Y , even if it is by a small amount.

Part d

Suppose you see that the ball is green. What is the probability that the coin turned up tails?

Let's solve this in terms of X and Y . Thus, we want $P(X = 0|Y = 1)$, which is easily solved using Bayes' Theorem:

$$\begin{aligned}P(X = 0|Y = 1) &= \frac{P(Y = 1|X = 0)P(X = 0)}{P(Y = 1|X = 0)P(X = 0) + P(Y = 1|X = 1)P(X = 1)} \\ &= \frac{(0.6)(0.5)}{(0.6)(0.5) + (0.4)(0.5)} \\ &= 0.6\end{aligned}$$

Problem 3

Let θ be a Beta(a, b) random variable, and suppose that the conditional distribution of X , given that θ , is Binomial(n, θ). Find the (marginal) probability mass function of X .

We can solve this by integrating out θ from the joint distribution.

$$\begin{aligned}
 f_X(X) &= \int P(X = x|\theta)P(\theta)d\theta \\
 &= \int \binom{n}{x} \theta^x (1-\theta)^{n-x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} d\theta \\
 &= \int \binom{n}{x} \theta^{x+a-1} (1-\theta)^{n+b-x-1} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} d\theta \\
 &= \binom{n}{x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int \theta^{x+a-1} (1-\theta)^{n+b-x-1} d\theta \\
 &= \binom{n}{x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(x+a-1)\Gamma(b+n-x-1)}{\Gamma(a+b+n-2)} \int \theta^{x+a-1} (1-\theta)^{n+b-x-1} d\theta \frac{\Gamma(a+b+n-2)}{\Gamma(x+a-1)\Gamma(b+n-x-1)} \\
 &= \binom{n}{x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(x+a-1)\Gamma(b+n-x-1)}{\Gamma(a+b+n-2)} (1)
 \end{aligned}$$

Problem 4

Full conditionals: Let X, Y, Z be random variables with joint density (discrete or continuous) $p(x, y, z)$. Show that

- (a) $p(x|y, z) = f(x, z)$, i.e., $p(x|y, z)$ is a function of x and z ;
- (b) $p(y|x, z) \propto g(y, z)$, i.e., $p(y|x, z)$ is a function of y and z ;
- (c) X and Y are conditionally independent given Z .

Part a

$$\begin{aligned}
 p(x|y, z) &= \frac{p(x, y, z)}{p(y)p(z)} \\
 \implies p(x|y, z) &\propto p(x, y, z) \\
 \implies p(x|y, z) &\propto f(x, z)g(y, z)h(z)
 \end{aligned}$$

Note that the only part of this that depends on X is $f(x, z)$, which implies that $p(x|y, z) \propto f(x, z)$.

Part b

Let's solve this in the exact same way.

$$\begin{aligned}
 p(y|x, z) &= \frac{p(x, y, z)}{p(x)p(z)} \\
 \implies p(y|x, z) &\propto p(x, y, z) \\
 \implies p(y|x, z) &\propto f(x, z)g(y, z)h(z)
 \end{aligned}$$

Similarly, the only portion of $f(x, z)g(y, z)h(z)$ that depends on y is the term $g(y, z)$. Thus, $p(y|x, z) \propto g(y, z)$.

Part c

To be conditionally independent, we need to show that $p(x, y|z) \propto p(x|z)p(y|z)$.

Let's begin with the same approach that we followed in parts a and b. We'll obviously need to get a little fancier for this part, but let's first show that $p(x, y|z) = f(x, z)g(y, z)h(z)$:

$$\begin{aligned} p(y, x|z) &= \frac{p(x, y, z)}{p(z)} \\ \implies p(y, x|z) &\propto p(x, y, z) \\ \implies p(y, x|z) &\propto f(x, z)g(y, z)h(z) \\ \implies p(y, x|z) &\propto f(x, z)g(y, z) \end{aligned}$$

Next, we need to think about $p(x|z)$ and $p(y|z)$.

$$\begin{aligned} p(x|z) &= \frac{p(x, z)}{p(z)} \\ p(x, z) &\propto f(x, z)f(z) \\ \implies p(x|z) &\propto f(x, z) \\ p(y|z) &= \frac{p(y, z)}{p(z)} \\ p(y, z) &\propto g(y, z)f(z) \\ \implies p(y|z) &\propto g(y, z) \end{aligned}$$

Combining these results, we see that $p(x, y|z) \propto f(x, z)g(y, z) \implies p(x, y|z) \propto p(x|z)p(y|z)$. Thus, x and y are conditionally independent.

Part 5

PH 3.1

Part a

Assume $Y_1, \dots, Y_{100}$ are, conditional on θ , i.i.d. binary random variables with expectation θ . Write down the joint distribution of $\Pr(Y_1 = y_1, \dots, Y_{100} = y_{100} | \theta)$ in a compact form. Also write down the form of $\Pr(\sum_{i=1}^{100} Y_i = y | \theta)$.

$$\begin{aligned} f(y_1, \dots, y_{100} | \theta) &= \prod_{i=1}^{100} f(y_i | \theta) \\ &= \theta^{\sum_{i=1}^{100} y_i} (1 - \theta)^{n - \sum_{i=1}^{100} y_i} \\ P\left(\sum_{i=1}^{100} Y_i = y | \theta\right) &= \binom{n}{\sum y_i} \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i} \end{aligned}$$

Part b

For the moment, suppose you believed that $\theta \in \{0.0, 0.1, \dots, 0.9, 1.0\}$. Given that the results of the survey were $\sum Y_i = 57$, compute $\Pr(\sum Y_i = 57 | \theta)$ for each of these 11 values of θ and plot these probabilities as a function of θ .

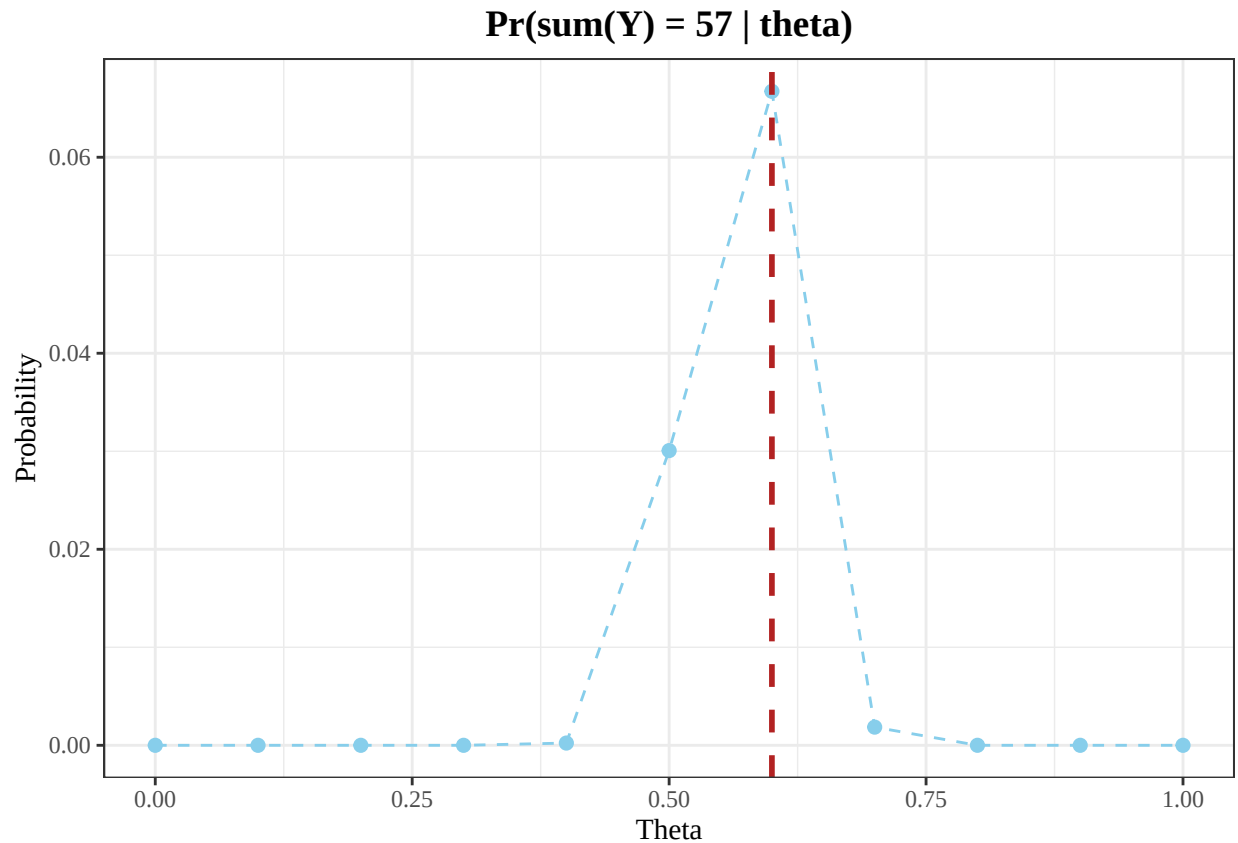
```
##### R #####
library(ggplot2)

# Define a function that will spit out the probability we're interested in
# computing
p_y_theta <- function(theta, n, y) {
  # y: a sum of bernouli trials
  # theta: a vector of values between 0 and 1 (can be length 1)
  # n: number of bernouli trials
  return(choose(n, y) * theta**y * (1 - theta)**(n - y))
}

# Define our primary input values (theta) and other given values (n, y)
theta <- seq(0, 1, length.out = 11)
n <- 100
y <- 57

# Figure out where the highest probability lies
max_prob_theta <- theta[
  which(p_y_theta(theta, n, y) == max(p_y_theta(theta, n, y)))
]

# Plot the outputs as nice and pretty as possible
plot_data <- data.frame(theta = theta, y = p_y_theta(theta, n, y))
ggplot(plot_data, aes(x = theta, y = y)) +
  geom_point(col = "skyblue", size = 2) +
  geom_line(col = "skyblue", lwd = 0.5, lty = 2) +
  geom_vline(xintercept = max_prob_theta, col = "firebrick", lwd = 1, lty = 2) +
  labs(title = "Pr(sum(Y) = 57 | theta)", x = "Theta", y = "Probability") +
  theme_bw() +
  theme(
    text = element_text(family = "serif"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )
)
```



The plot shows a maximum at 0.6.

Part c

Now suppose you originally had no prior information to believe one of these θ -values over another, and so $\Pr(\theta = 0.0) = \Pr(\theta = 0.1) = \dots = \Pr(\theta = 0.9) = \Pr(\theta = 1.0)$. Use Bayes' rule to compute $p(\theta | \sum Y_i = 57)$ for each θ -value. Make a plot of this posterior distribution as a function of θ .

I want to note that I used generative AI to assist me with the following code for this problem. I derived the the math to implement, and ChatGPT assisted with the coding:

```
##### R #####

# Define the function to compute posterior probability using Bayes' theorem
p_theta_bayes <- function(theta, n, y) {
  # Compute P(Y = 57 | theta) for each
  likelihood <- choose(n, y) * theta^y * (1 - theta)^(n - y)

  # Prior probabilities (uniform over 11 values)
  prior <- rep(1 / length(theta), length(theta))

  # Compute P(Y = 57) as the sum of likelihood * prior over all
  p_y <- sum(likelihood * prior)

  # Compute posterior using Bayes' rule
  posterior <- (likelihood * prior) / p_y
}
```

```

    return(posterior)
}

# Define theta values and known values of n and y
theta <- seq(0, 1, length.out = 11)
n <- 100
y <- 57

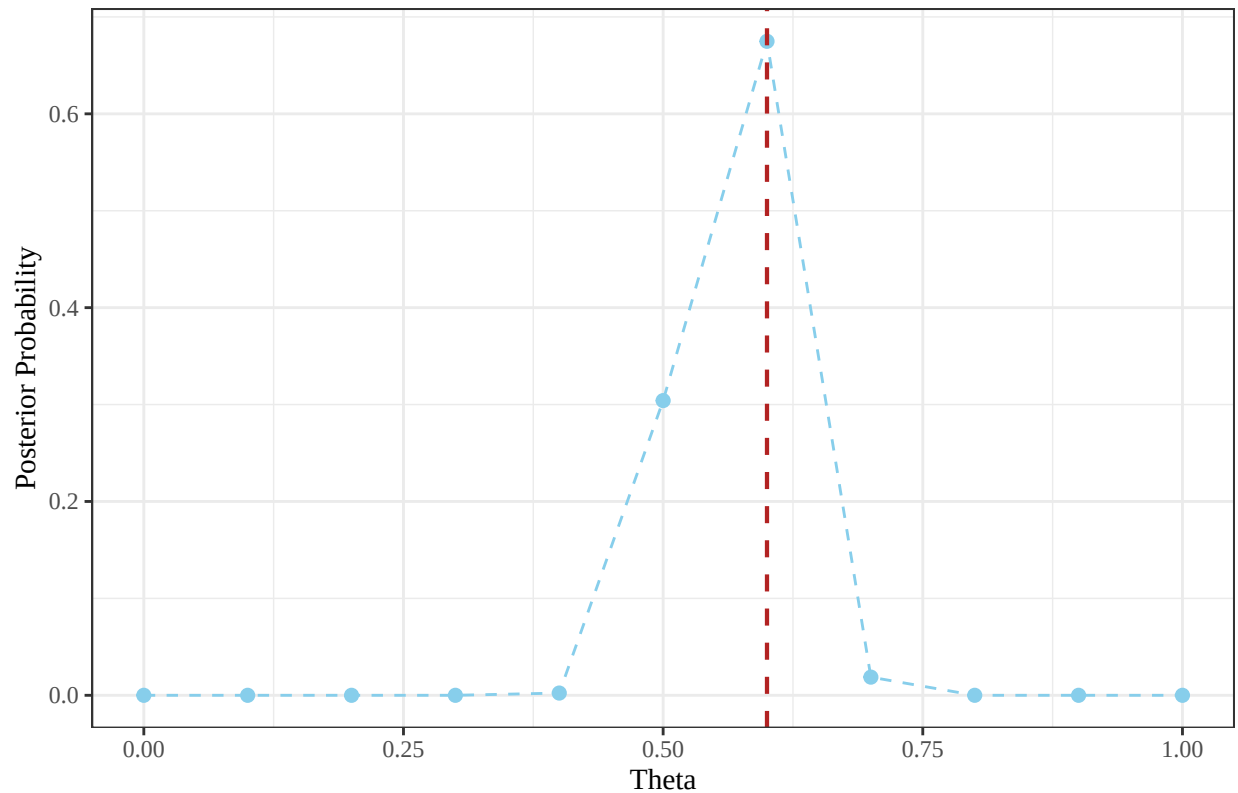
# Compute the posterior probabilities
posterior_probs <- p_theta_bayes(theta, n, y)
max_prob_theta <- theta[
  which(posterior_probs == max(posterior_probs))
]

# Plot the posterior distribution
plot_data <- data.frame(theta = theta, posterior = posterior_probs)

ggplot(plot_data, aes(x = theta, y = posterior)) +
  geom_point(col = "skyblue", size = 2) +
  geom_line(col = "skyblue", lwd = 0.5, lty = 2) +
  geom_vline(col = "firebrick", lwd = 0.75, lty = 2, xintercept = max_prob_theta) +
  labs(
    title = "Posterior Probability Distribution P( | Y = 57)",
    x = "Theta", y = "Posterior Probability"
  ) +
  theme_bw() +
  theme(
    text = element_text(family = "serif"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )

```

Posterior Probability Distribution $P(\cdot | Y = 57)$



Max at $\theta = 0.6$.

Part d

Now suppose you allow θ to be any value in the interval $[0, 1]$. Using the uniform prior density for θ , so that $p(\theta) = 1$, plot the posterior density $p(\theta) \times \Pr(\sum Y_i = 57 | \theta)$ as a function of θ .

```
##### R #####
library(ggplot2)

# Define function to compute posterior density
posterior_density <- function(theta_vals, n, y) {
  # Compute likelihood  $P(Y = 57 | \theta)$ 
  likelihood <- choose(n, y) * theta_vals^y * (1 - theta_vals)^(n - y)

  # Since prior is uniform (1), posterior is proportional to likelihood
  posterior <- likelihood

  return(posterior)
}

# Define a continuous range of theta values
theta_vals <- seq(0, 1, length.out = 1000) # More points for smooth curve
n <- 100
y <- 57

# Compute posterior density
```



```

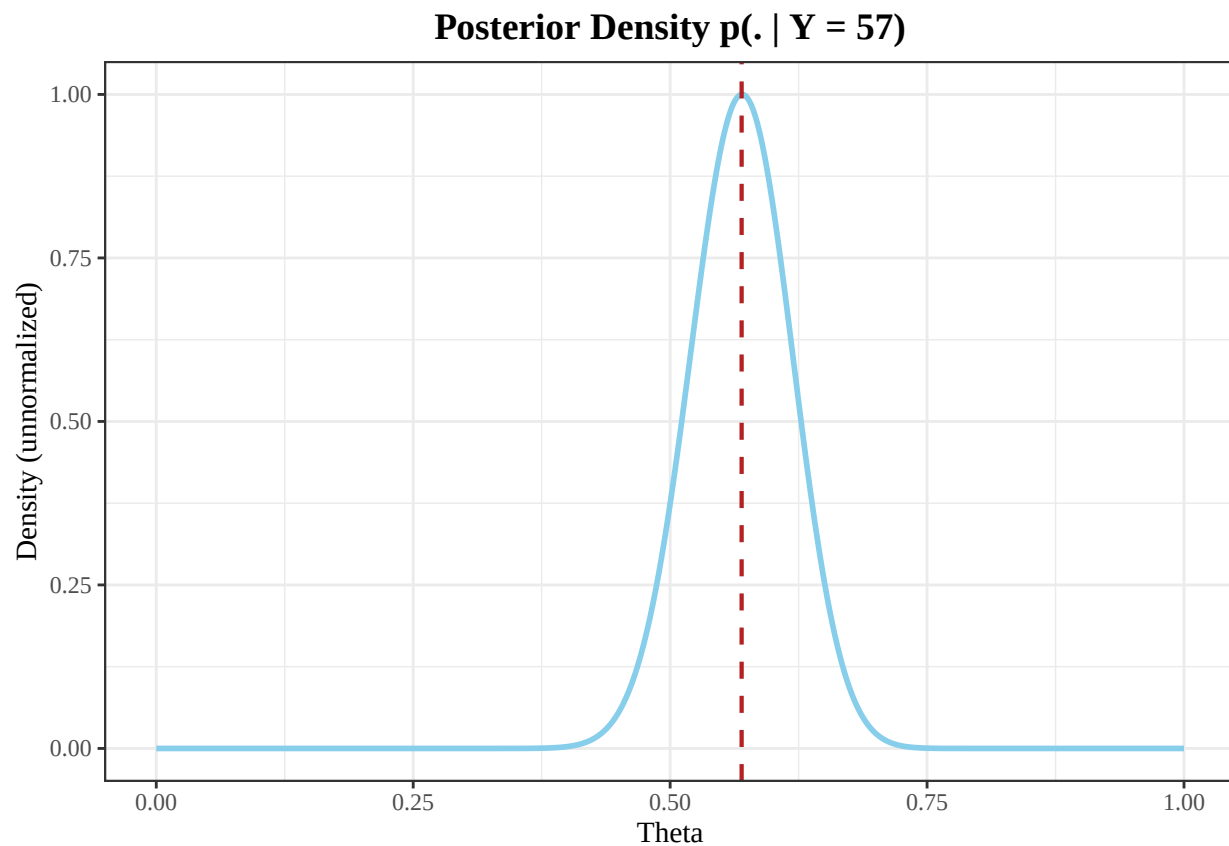
posterior_vals <- posterior_density(theta_vals, n, y)
max_prob_theta <- theta_vals[
  which(posterior_vals == max(posterior_vals))
]

# Normalize for better visualization (optional, but makes y-axis interpretable)
posterior_vals <- posterior_vals / max(posterior_vals)

# Plot the posterior density
plot_data <- data.frame(theta = theta_vals, posterior = posterior_vals)

ggplot(plot_data, aes(x = theta, y = posterior)) +
  geom_line(col = "skyblue", lwd = 1) +
  geom_vline(col = "firebrick", lwd = 0.75, lty = 2, xintercept = max_prob_theta) +
  labs(
    title = "Posterior Density p( | Y = 57)",
    x = "Theta", y = "Density (unnormalized)"
  ) +
  theme_bw() +
  theme(
    text = element_text(family = "serif"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )

```



Max at $\theta = 0.5695696$.

Part e

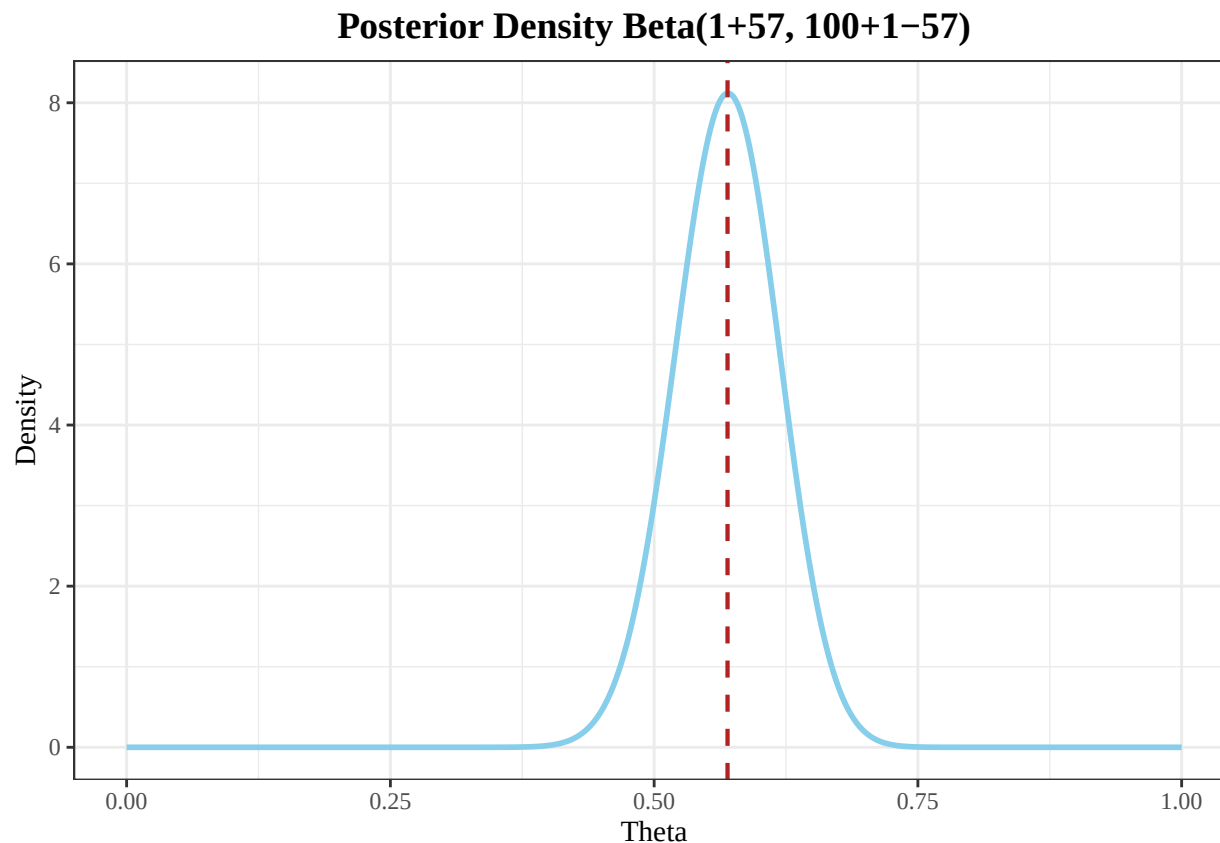
```
##### R #####

# Define a function to return a posterior Beta(1+57, 1+100-57) density given
# an input of theta values
posterior_beta_density <- function(theta_vals) {
  return(dbeta(theta_vals, 1 + 57, 1 + 100 - 57))
}

# Get a set of density values like we did in part d
posterior_probs <- posterior_beta_density(theta_vals)
max_prob_theta <- theta_vals[
  which(posterior_probs == max(posterior_probs))
]

# Combine into a dataframe
plot_data <- data.frame(theta = theta_vals, density = posterior_probs)

# Now let's get back to plotting!
ggplot(plot_data, aes(x = theta, y = density)) +
  geom_line(col = "skyblue", lwd = 1) +
  geom_vline(col = "firebrick", lwd = 0.75, lty = 2, xintercept = max_prob_theta) +
  labs(
    title = "Posterior Density Beta(1+57, 100+1-57)",
    x = "Theta", y = "Density"
  ) +
  theme_bw() +
  theme(
    text = element_text(family = "serif"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )
```



Max at $\theta = 0.5695696$.

I'm honestly pretty surprised that they have all given the same (roughly) optimal values of θ . My hunch is that if I had made the first two plots as granular as the second two, I would see this same θ value of 0.569. We've used four different methods and seem to get the same results consistently.

Part 6

PH 3.2

Sensitivity analysis: It is sometimes useful to express the parameters a and b in a beta distribution in terms of $\theta = a/(a + b)$ and $n_0 = a + b$, so that $a = \theta n_0$ and $b = (1 - \theta)n_0$. Reconsidering the sample survey data in Exercise 3.1, for each combination of $\theta \in \{0.1, 0.2, \dots, 0.9\}$ and $n_0 \in \{1, 2, 8, 16, 32\}$ find the corresponding a, b values and compute $\Pr(\hat{\theta} > 0.5 | \sum Y_i = 57)$ using a $\text{beta}(a, b)$ prior distribution for θ . Display the results with a contour plot, and discuss how the plot could be used to explain to someone whether or not they should believe that $\theta > 0.5$, based on the data that $\sum Y_i = 57$.

```
##### R #####
```

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union

# Define prior values
theta0_values <- seq(0.1, 0.9, by = 0.1)
n0_values <- c(1, 2, 8, 16, 32)

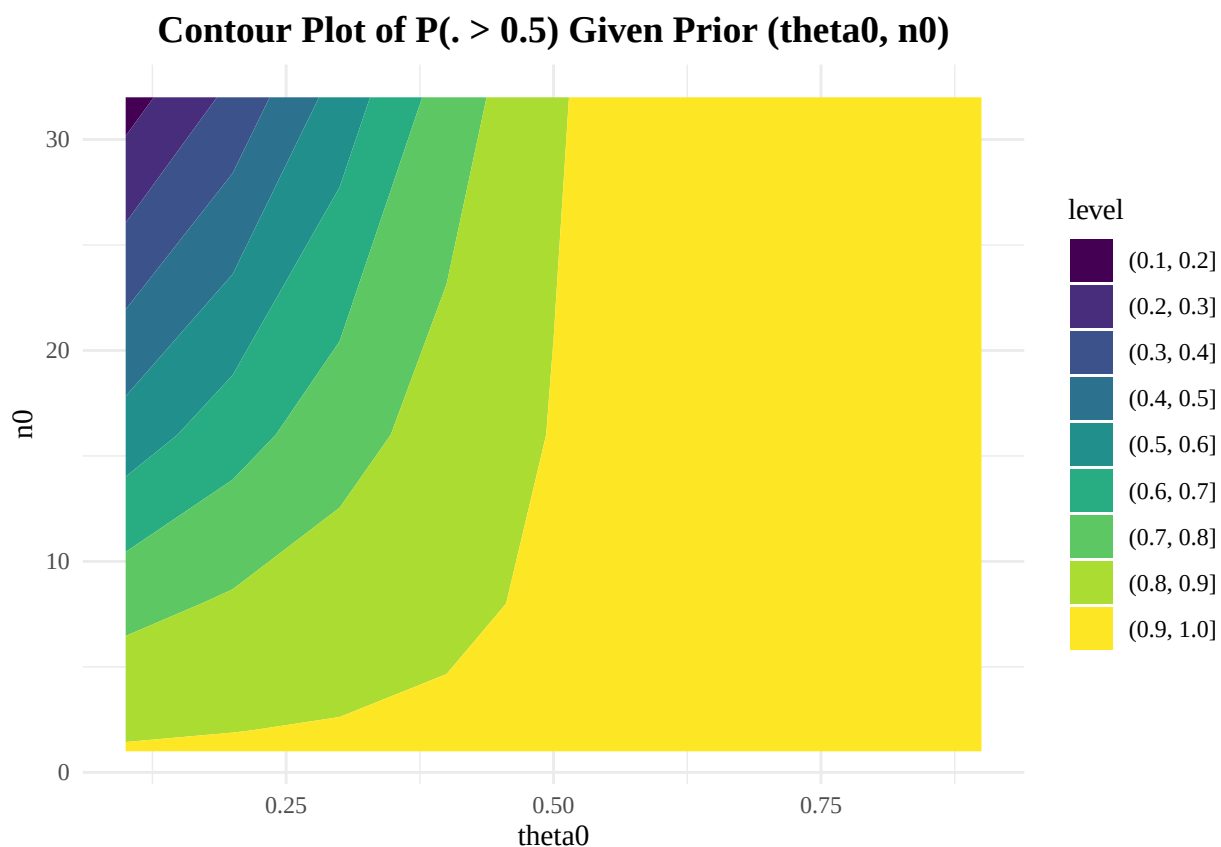
# Create an empty dataframe to store results
results <- expand_grid(theta0 = theta0_values, n0 = n0_values)

# Observed data
y_sum <- 57
n <- 100

# Compute probability  $P(> 0.5)$  for each combination
results <- results %>%
  rowwise() %>%
  mutate(
    a = theta0 * n0,
    b = (1 - theta0) * n0,
    a_post = a + y_sum,
    b_post = b + (n - y_sum),
    prob_theta_greater_05 = 1 - pbeta(0.5, a_post, b_post)
  )

# Plot using ggplot2
ggplot(results, aes(x = theta0, y = n0, z = prob_theta_greater_05)) +
  geom_contour_filled() +
  labs(
    title = "Contour Plot of  $P(> 0.5)$  Given Prior ( $\theta_0$ ,  $n_0$ )",
    x = "theta0",
    y = "n0"
  ) +
  theme_minimal() +
  theme(
    text = element_text(family = "serif"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )

```



The contour plot shows the probability that θ is greater than 0.5 given the data. With higher values of θ_0 , we see that our posterior probabilities are also higher. However, with lower values of θ_0, n_0 , which represents our prior certainty, is inversely related. That is, if I'm much more certain that $\theta_0 < 0.5$, then my posterior probability that $\theta > 0.5$ is lower. This should be intuitive and matches what we see on the plot.